CloudSense AI Project Abstract

ABSTRACT

Modern cloud-native environments frequently experience significant financial inefficiencies due to over-provisioned resources, idle virtual machines, unattached storage volumes, obsolete snapshots, and the absence of intelligent cloud cost management. In addition to operational waste, organizations are increasingly exposed to economic security threats such as Denial-of-Wallet (DoW) attacks, where malicious workloads intentionally trigger excessive cloud resource consumption, resulting in unnecessary financial expenditure. Existing FinOps platforms primarily provide historical billing visualization and rule-based monitoring but offer limited capabilities for predictive analytics, business-aware decision support, explainable recommendations, and secure cloud remediation.

This project proposes CloudSense AI: An Intelligent Cloud Resource and Cost Optimization Framework using Machine Learning, Explainable Artificial Intelligence (XAI), Model Context Protocol (MCP), and AWS SDK (Boto3). The framework integrates Prophet for cloud cost forecasting, XGBoost for business-aware Unit Economics prediction, and Isolation Forest for cloud anomaly and Denial-of-Wallet detection. Explainable AI provides transparent recommendations with confidence scores, estimated savings, risk assessment, feature importance, and root-cause analysis to support engineering decisions.

To reduce manual operational effort, the framework incorporates an Intelligent Cloud Remediation Engine that leverages MCP for standardized communication with enterprise cloud services and AWS SDK (Boto3) for secure execution of approved cloud optimization actions. Instead of generating Infrastructure-as-Code templates, the framework performs controlled remediation tasks such as managing EC2 instances, EBS volumes, snapshots, Auto Scaling Groups, and other AWS resources through secure API-based automation. All remediation actions follow a Human-in-the-Loop (HITL) workflow requiring administrator approval before execution.

The framework also provides real-time notifications through Slack, Microsoft Teams, Discord, Email, and custom webhooks. An AI-powered conversational assistant enables users to interact with cloud infrastructure using natural language and receive contextual explanations of AI recommendations.

The proposed framework is expected to improve cloud resource utilization, accurately forecast cloud expenditure, detect cloud cost anomalies and Denial-of-Wallet attacks, reduce manual cloud management effort, strengthen cloud governance through secure AI-assisted remediation, and enhance operational transparency, financial efficiency, and decision-making in enterprise cloud environments.

Keywords: Machine Learning (ML), Explainable Artificial Intelligence (XAI), Cloud Computing, Cloud Resource Optimization, Cloud Cost Optimization, FinOps, Model Context Protocol (MCP), AWS SDK (Boto3), Intelligent Cloud Remediation, Denial-of-Wallet (DoW) Detection, Unit Economics, Human-in-the-Loop (HITL).